

Here is the final code from the MiniZinc sample set with comments and the output procedure removed for brevity:

```
int: n;
array [1..n] of var 1..n: q;
include "alldifferent.mzn";
constraint alldifferent(q);
constraint alldifferent([ q[i] + i | i in 1..n]);
constraint alldifferent([ q[i] - i | i in 1..n]);
solve :: int_search(q, first_fail, indomain_min, complete)
satisfy;
```

2	3	4	5	0	-1	-2	-3
3	4	5	6	1	0	-1	-2
4	5	6	7	2	1	0	-1
5	6	7	8	3	2	1	0

The above tables illustrate how sum of row and column number (left) and their difference (right) vary.

The final solve line in the code above is actually quite magical. The parameters provided there can be the difference between a solution derived in seconds, or in hours, or even not at all.

Let's see what they do then. First of all, `int_search` as you may guess from the name, searches for integer values within `q`. The `first_fail` parameter is interesting as it tells the compiler to first try the solutions that have a greater chance of failing first. In context to the `n-Queens` program, the solver will try to place a queen in the column with the fewest free spaces. This way, if it fails, it will quickly backtrack rather than going down the wrong path for longer. The third option – `indomain_min` instructs the compiler to start with the lowest row number (like we did in the manual solution above). Finally, the `complete` parameter makes the program look for all possible combinations till it finds a solution.

As you can see, while the solution is short, sweet and fast, it takes considerable understanding of the solution to model the problem in a way that can lead to a solution. Like any programming language, this is an intuition that the programmer needs to develop, as constraint programming has no shortcuts.

Conclusion:

Constraint programming is an exciting paradigm of computing, one that many people don't come across soon enough. As you might have guessed from the above article though, it is quite powerful when applied correctly.

You might also be able to understand why it is not more commonly used. It isn't really something you can write an entire software package in. For some kind of problems though, it can be the perfect tool for the job.

The good news here is that, you don't need to use a pure constraint programming language to take advantage of this paradigm. For one, it is possible to embed such programs within traditional software so the constraint programming bits can be written in their native language. For instance, you could build a Sudoku solver that combines a Python / Java / C++ program for the UI with a constraint programming at the back end that actually solves the problem.

The popular ECLiPSe Constraint Programming System (not to be confused with the Eclipse IDE) can be embedded and called from C / C++ and has an interface for C / C++, Python, Java, and even PHP. The various Prolog implementations also often have a C / C++ interface.

There are also many libraries available that add constraint programming features to traditionally imperative programming languages. For users of C/C++, there are Gecode, Minion and Comet. For Java developers there are Choco and Cream. For Python, you have Python-constraint and Numberjack. There is also a multi-language constraint programming library by Google called `or-tools` that supports C++, Python, Java and .NET.

There are many domains and paradigms of programming that can make certain kinds of programs much simpler to develop. Now that you have some knowledge of constraint programming in your vocabulary to pull out whenever needed, hopefully you will explore and find out more. >

*n-links solution



*ECLiPSe

>>If you're looking for a programming environment that focuses on constraint programming, ECLiPSe is an open source project to look into. The ECLiPSe programming language is based on and compatible with Prolog.

<http://dgit.in/EclipSe>



*Numberjack

>>Looking for a constraint programming library for Python? Look no further than Numberjack. This free and open source library comes with a number of solvers and is easy to use.

<http://dgit.in/Numberjack>



*or-tools

>>Google has its own set of tools and libraries for constraint programming called `or-tools`. This library is open source, and can work with C++, Python, Java, and C#. So what are you waiting for?

<http://dgit.in/or-tools>